ARCH - Student Portal

Phase 1

Team 1

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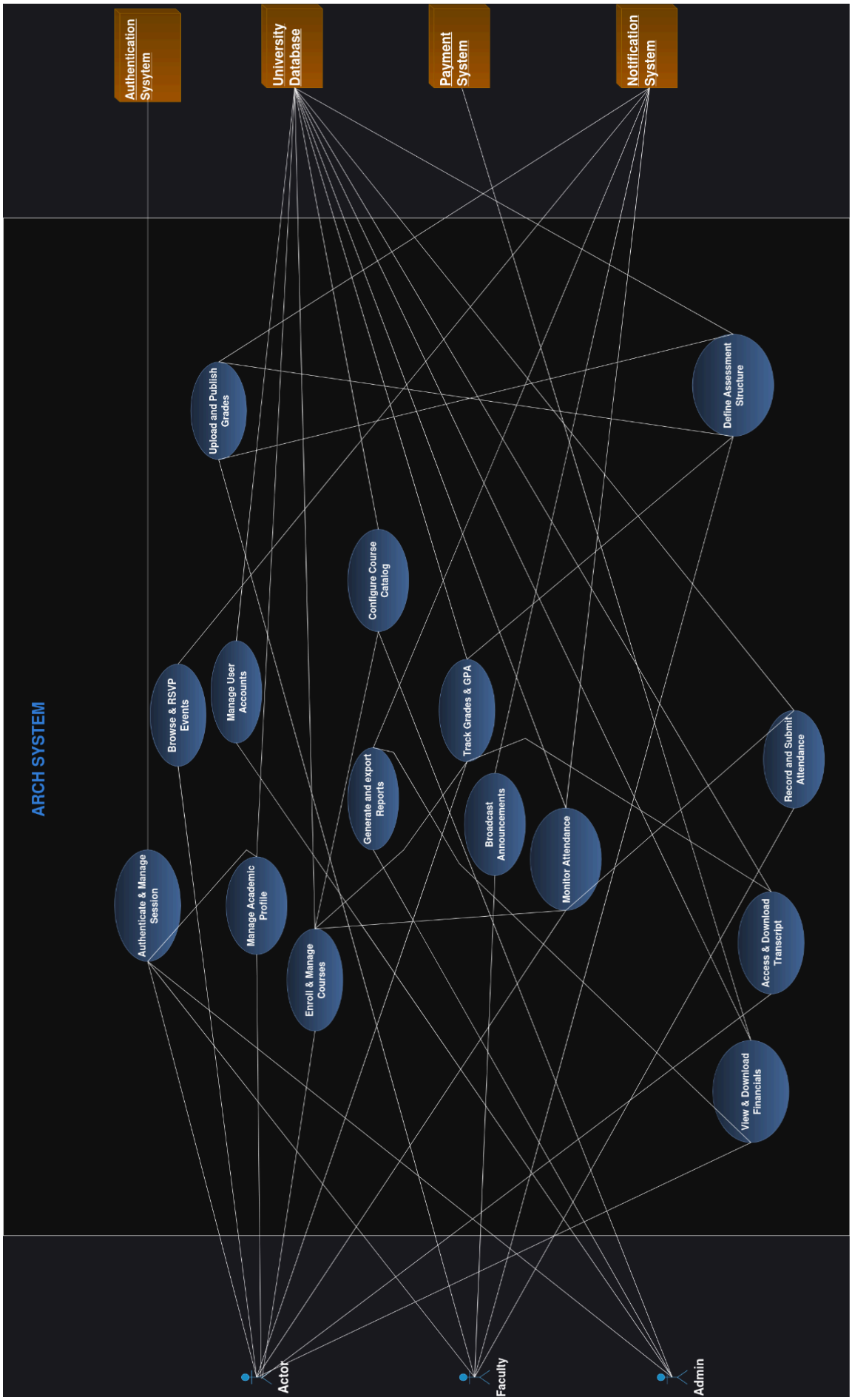
Table of Contents

1. Use Case Diagram
2. Use Case Description 1
3. Use Case Description 2
4. Use Case Description 3
5. Use Case Description 4
6. Use Case Description 5
7. Use Case Description 6
8. Use Case Description 7
9. Use Case Description 8
10. Use Case Description 9
11. Use Case Description 10
12. Use Case Description 11
13. Use Case Description 12
14. Use Case Description 13
15. Use Case Description 14
16. Use Case Description 15
17. Use Case Description 16
18. Use Case Description 17
19. Use Case Description 18
20. Use Case Description 19
21. Use Case Description 20
22. Use Case Description 21
23. Use Case Description 22
24. Use Case Description 23
25. Use Case Description 24
26. Analysis Class Diagram

USE CASE DIAGRAM



Identifier	**UC-01**
Name	**Student Login**
Summary	**A student logs into the ARCH portal using their credentials**
Priority	**High**
Actors	**Student, Authentication System**
Dependencies	**N/A (As entry point)**
Pre-condition(s)	**Student has a registered account with valid credentials**
Post-condition(s)	**Student is authenticated and redirected to their personal dashboard**

Typical Course of Action		
S #	**Actor Action**	**System Response**
1	**Student navigates to the ARCH login page**	**System displays the login form**
2	**Student enters their ID and password**	**System validates the input fields**
3	**Student clicks the Login button**	**System authenticates credentials**
4		**System redirects to the Student Dashboard**

Alternate Course of Action		
S #	**Actor Action**	**System Response**
1	**Student enters incorrect**	**System displays an error**

	credentials	**message**

Identifier	**UC-02**
Name	**Teacher Login**
Summary	**An Teacher logs into the ARCH portal**
Priority	**High**
Actors	**Teacher**
Dependencies	**N/A (As entry point)**
Pre-condition(s)	**Teacher has a registered account**
Post-condition(s)	**Admin is authenticated and redirected**

Typical Course of Action		
S #	**Actor Action**	**System Response**
1	**Teacher navigates to the ARCH login page**	**System displays the login form**
2	**Teacher enters their ID and password**	**System validates the input fields**
3	**Teacher clicks the Login button**	**System authenticates credentials**
4		**System redirects to the Teacher Dashboard**
Alternate Course of Action		
S #	**Actor Action**	**System Response**
1	**Student enters incorrect**	**System displays an error**

	credentials	**message**

Identifier	**UC-03**
Name	**Student Logout**
Summary	**A logged-in student ends their session**
Priority	**Medium**
Actors	**Student**
Dependencies	**User must be authenticated first**
Pre-condition(s)	**Student is currently logged in**
Post-condition(s)	**Student session is terminated**

Typical Course of Action		
S #	**Actor Action**	**System Response**
1	**Student clicks the Logout button**	**System prompts a confirmation dialog**
2	**Student confirms logout action**	**System terminates the active session**
3		**System redirects the student to the login page**
Alternate Course of Action		

S #	Actor Action	System Response
1	Student closes the browser without logging out	System automatically expires the session
2	Student cancels the logout action	System dismisses the dialog and keeps the session active

Identifier	UC-04
Name	Role Based Access
Summary	The system restricts or grants access to features and pages based on user's role
Priority	High
Actors	Student, Teacher, Database
Dependencies	Depends on login role
Pre-condition(s)	User is successfully authenticated
Post-condition(s)	User can only access features and pages permitted for their assigned role

Typical Course of Action

S #	Actor Action	System Response
1	User logs in successfully	System retrieves the user's role
2	User attempts to navigate to a page	System grants access and renders the page if the role

		matches
Alternate Course of Action		
S #	**Actor Action**	**System Response**
1	**User gains access to an unrestricted page**	**System checks role and redirects to home page**

Identifier	**UC-05**
Name	**View Cumulative GPA**
Summary	**A student views their overall Cumulative GPA calculated across all completed semesters**
Priority	**High**
Actors	**Student Database**
Dependencies	**Student must be logged in**
Pre-condition(s)	**Student has >= 1 Completed Semester**
Post-condition(s)	**Student's Cumulative GPA is displayed on the Academic Overview dashboard**

Typical Course of Action		
S #	**Actor Action**	**System Response**
1	**Student navigates to the**	**System fetches academic**

	Academic Overview section	**records**
2		**System calculates and displays Cumulative GPA**
Alternate Course of Action		
S #	**Actor Action**	**System Response**
1	**Student has no completed semesters**	**Display appropriate message instead**

Identifier	**UC-06**
Name	**View Semester GPA**
Summary	**A student views their GPA for a specific semester**
Priority	**High**
Actors	**Student, Database**
Dependencies	**Student must be enrolled in a program**
Pre-condition(s)	**Student has completed at least one semester**
Post-condition(s)	**The selected semester's GPA is displayed to the student**

Typical Course of Action		
S #	**Actor Action**	**System Response**

1	**Student navigates to the Academic Overview**	**System loads the list of available semesters**
2		**System displays Semester GPA with specific semester**
Alternate Course of Action		
S #	**Actor Action**	**System Response**
1	**Student has no completed semesters**	**Display appropriate message instead**

Identifier	**UC-07**
Name	**View Credit Progress**
Summary	**A student views their overall credit hour progress, showing completed, in-progress, and remaining credits toward their degree**
Priority	**High**
Actors	**Student, Database**
Dependencies	**Depends on Role Based Access i.e. the student must be enrolled in a program**
Pre-condition(s)	**Student is enrolled in a Program**
Post-condition(s)	**Student's credit progress is displayed on the Academic Overview**

Typical Course of Action		
S #	**Actor Action**	**System Response**
1	**Student navigates to the Academic Overview section**	**System fetches the student's enrollment and credit data**
2		**System calculates completed, in-progress, and remaining credit hours**
Alternate Course of Action		
S #	**Actor Action**	**System Response**
1	**Student has not yet completed any credits**	**System displays 0 credits**

Identifier	**UC-08**
Name	**Enroll in a Course**
Summary	**A student enrolls in an available course**
Priority	**High**
Actors	**Student, Database**
Dependencies	**Depends on View Credit Progress i.e. the student must be eligible based on credit standing**
Pre-condition(s)	**Student is logged in and the course has available seats**

Post-condition(s)	Student is successfully enrolled	
Typical Course of Action		
S #	**Actor Action**	**System Response**
1	Student navigates to the Course Registration section	System displays the list of available courses
2	Student selects a course to enroll in	System checks seat availability and prerequisite eligibility
3	Student confirms the enrollment	System registers the student in the course
Alternate Course of Action		
S #	**Actor Action**	**System Response**
1	Selected course has no available seats	System displays a message
2	Student has not met prerequisites	System blocks enrollment
3	Enrollment window is closed	System displays a message

Identifier	**UC-09**
Name	**Unenroll from a Course**
Summary	**A student removes themselves from a currently enrolled course**
Priority	**High**
Actors	**Student, Database**
Dependencies	**Depends on Enroll in a Course i.e. the student must**

	already be enrolled
Pre-condition(s)	**Student is enrolled in course and drop period is open**
Post-condition(s)	**Student is removed from the selected course**

Typical Course of Action		
S #	**Actor Action**	**System Response**
	Student navigates to the Course Registration section	**System displays currently enrolled courses**
	Student selects the course they wish to drop	**System displays course info and drop prompt**
	Student confirms	**System removes the student**

Alternate Course of Action		
S #	**Actor Action**	**System Response**
	Drop period is closed	**Appropriate message is displayed**
	Minimum credit limit violated	**System blocks action**
	Student cancels unenrollment	**Enrollment intact**

Identifier	**UC-10**
Name	**Shift Course Section**
Summary	**A student switches from one section of an enrolled course to another available section**
Priority	**Low**

Actors	**Student, Database**
Dependencies	**Depends on Enroll in a Course i.e. the student must already be enrolled in the course**
Pre-condition(s)	**Student is enrolled in the course and alternate section has available seats**
Post-condition(s)	**Student is shifted to the new section**

Typical Course of Action		
S #	**Actor Action**	**System Response**
1	**Student navigates to the Course Registration**	**System displays the student's currently enrolled courses**
2	**Student selects the option to shift section**	**System displays all available sections**
3	**Student selects the desired section**	**System checks seat availability and changes section**

Alternate Course of Action		
S #	**Actor Action**	**System Response**
1	**Desired section has no available seats**	**System displays "Selected section is full**
2	**New section has a timetable clash**	**System alerts the student of the conflict and blocks shift**

Identifier	**UC-11**
Name	**View Announcements**
Summary	**A student views all official announcements**

	broadcast by the university administration
Priority	**Low**
Actors	**Student**
Dependencies	**Depends on Role Based Access i.e. the student must be logged in**
Pre-condition(s)	**One University Wide Announcement exists**
Post-condition(s)	**All active university-wide announcements are displayed to the student in the News Feed**

Typical Course of Action		
S #	**Actor Action**	**System Response**
1	**Student navigates to the Centralized Notices**	**System retrieves all active university-wide announcements**
2		**System displays the announcements**
3	**Student clicks on an announcement**	**System displays the full details of the selected announcement**

Alternate Course of Action		
S #	**Actor Action**	**System Response**
1	**No active announcements**	**System displays appropriate message**

Identifier	**UC-12**

Name	**Create Faculty Alerts**
Summary	**Teacher creates course specific alerts**
Priority	**Medium**
Actors	**Teacher, Database**
Dependencies	**Depends on Role Based Access i.e. the teacher must be logged in**
Pre-condition(s)	**Teacher is logged in**
Post-condition(s)	**Faculty Alert is displayed for respective section and course**

Typical Course of Action

S #	Actor Action	System Response
1	Teacher navigates to centralised feed	System fetches all details
2	Teacher creates alert	System takes all details and sends it out

Alternate Course of Action

S #	Actor Action	System Response
1	Teacher cancels mid-creation	System takes back to previous tab

Identifier	**UC-13**
Name	**View Faculty Alerts**
Summary	**A student views course-specific alerts and notifications**
Priority	**Medium**
Actors	**Student, Database**
Dependencies	**Depends on Create Faculty Alerts i.e. at least one alert must have been posted by faculty**
Pre-condition(s)	**Student is enrolled in at least one course, and the respective faculty has posted an alert**
Post-condition(s)	**All faculty alerts for the student are displayed**

Typical Course of Action

S #	Actor Action	System Response
1	Student navigates to the Centralized Notices	System fetches all faculty alerts linked to the student's enrolled courses
2		System displays alerts
3	Student clicks on an alert	System displays the full details of the selected faculty alert

Alternate Course of Action

S #	Actor Action	System Response
1	No faculty alerts have been posted	System displays appropriate message

2	**Student is not enrolled in any courses**	**System displays appropriate message**

Identifier	**UC-14**
Name	**Update Student's Personal Info**
Summary	**A student updates their personal details**
Priority	**Medium**
Actors	**Student, Database**
Dependencies	**Depends on Student Login i.e. the student must have an existing profile**
Pre-condition(s)	**Student has an existing profile in the system**
Post-condition(s)	**Student's new information is saved and updated**

Typical Course of Action		
S #	**Actor Action**	**System Response**
1	**Student navigates to the Profile Management section**	**System displays the student's current information**
2	**Student edits the desired fields**	**System validates the format of the entered data**
3	**Student clicks "Save Changes"**	**System updates data in database**

Alternate Course of Action		
S #	**Actor Action**	**System Response**
1	**Student enters data in an invalid**	**System displays error message**

	format	

Identifier	**UC-15**
Name	**Update Teacher's Info**
Summary	**A Teacher updates their personal details**
Priority	**Medium**
Actors	**Teacher, Database**
Dependencies	**Depends on Teacher Login i.e. the teacher must have an existing profile**
Pre-condition(s)	**Teacher has an existing profile in the system**
Post-condition(s)	**Teacher's new information is saved and updated**

S #	**Actor Action**	**System Response**
1	**Teacher goes to the Profile Management section**	**System displays the Teacher's current information**
2	**Teacher edits the desired fields**	**System validates the format of the entered data**
3	**Teacher clicks "Save Changes"**	**System updates data in database**
Alternate Course of Action		
S #	**Actor Action**	**System Response**

1	**Teacher enters data in an invalid format**	**System displays error message**

Identifier	**UC-16**
Name	**View Section Timetable**
Summary	**A student views their weekly class schedule organised according to the sections they are in**
Priority	**High**
Actors	**Student, Database**
Dependencies	**Depends on Enroll in a Course i.e. the student must be in at least 1 section**
Pre-condition(s)	**Student is enrolled in at least 1 course section for the current semester**
Post-condition(s)	**Student's section wise timetable is shown with details inclusive of days, class times and room details**

Typical Course of Action

S #	Actor Action	System Response
1	**Student navigates to the Timetable section**	**System fetches the student's enrolled course sections**
2		**System organizes and displays the schedule**

Alternate Course of Action		
S #	**Actor Action**	**System Response**
1	**Student is not enrolled in any courses**	**System displays appropriate message**

Identifier	**UC-17**
Name	**View Course Grades**
Summary	**A student views the grades they have received for each course**
Priority	**High**
Actors	**Student, database**
Dependencies	**Depends on Enroll in a Course i.e. the student must have at least one finalized grade**
Pre-condition(s)	**Student has at least one course with a finalised grade**
Post-condition(s)	**A list of courses along with their respective grades is displayed**

Typical Course of Action		
S #	**Actor Action**	**System Response**
1	**Student moves to the Grade Transcript section**	**System fetches grades from Database**

2		**System displays a list of courses with their letter grades and grade points**
Alternate Course of Action		
S #	**Actor Action**	**System Response**
1	**Student has no completed courses**	**System displays appropriate error message**

Identifier	**UC-18**
Name	**View Credit Weightage to GPA**
Summary	**A student views how each course's credit hours contribute to and affect their overall GPA**
Priority	**Medium**
Actors	**Student, Database**
Dependencies	**Depends on View Course Grades i.e. graded course records must exist**
Pre-condition(s)	**Student has at least 1 weighted graded course**
Post-condition(s)	**Each course's credit hours and their weighted contribution to the GPA is displayed**

Typical Course of Action		
S #	**Actor Action**	**System Response**

1	**Student navigates to the Grade Transcript section**	**System fetches the student's grade and credit hour records**
2		**System calculates the weighted contribution of each course to the overall GPA**
3		**System displays course name, credit hours, grade, and GPA contribution per course**
Alternate Course of Action		
S #	**Actor Action**	**System Response**
1	**Student has no graded courses on record**	**System displays appropriate message**

Identifier	**UC-19**
Name	**View Marks Breakdown**
Summary	**A student views their marks and performance breakdown for each individual assessment**
Priority	**High**
Actors	**Student, Database**
Dependencies	**Depends on Enroll in a Course i.e. at least one assessment result must have been entered**
Pre-condition(s)	**Student is enrolled in at least one course, and at least one assessment result has been entered**
Post-condition(s)	**A breakdown per assessment is displayed**
Typical Course of Action	

S #	Actor Action	System Response
1	Student checks Marks section	System fetches all enrolled courses
2	Student selects a course to view	System retrieves the list of assessments and scores
3		System displays each assessment with its details
Alternate Course of Action		
S #	Actor Action	System Response
1	No Graded Assessments present	Relevant error message is displayed
2	Student is not enrolled in any course	System displays enrollment prompt

Identifier	UC-20
Name	View Attended Lecture Count
Summary	A student views the total number of lectures they have attended for each of their enrolled courses
Priority	High
Actors	Student, Database
Dependencies	Depends on Enroll in a Course i.e. attendance must have been recorded for at least one lecture
Pre-condition(s)	Student is enrolled in >= 1 course and has attendance recorded

Post-condition(s)	Total number of attended lectures is displayed	
Typical Course of Action		
S #	**Actor Action**	**System Response**
1	**Student goes to Attendance Section**	**System fetches enrolled courses**
2	**Student selects course to view**	**System retrieves the attendance data for that specific course**
3		**System displays lectures held/lectures attended**
Alternate Course of Action		
S #	**Actor Action**	**System Response**
1	**Student not enrolled in any courses**	**System displays enrollment prompt**
2	**No attendance recorded yet**	**Relevant message displayed**

Identifier	**UC-21**
Name	**View Minimum Required Lectures**
Summary	**A student views the minimum number of lectures they must attend**
Priority	**High**
Actors	**Student, Database**
Dependencies	**Depends on View Attended Lecture Count i.e. attendance data must be available**

Pre-condition(s)	**Student is enrolled in >= 1 course**	
Post-condition(s)	**Minimum lecture count and current count displayed**	
Typical Course of Action		
S #	**Actor Action**	**System Response**
1	**Student goes to Attendance Section**	**System fetches enrolled courses**
2	**Student selects course to view**	**System retrieves the attendance data for that specific course**
3		**System displays minimum required/attended lectures**
Alternate Course of Action		
S #	**Actor Action**	**System Response**
1	**No lectures have been held yet**	**System displays relevant error message**

Identifier	**UC-22**
Name	**View Tardies**
Summary	**A student views the number of times they have been marked late or tardy**
Priority	**Low**
Actors	**Student, Database**

Dependencies	**Depends on View Attended Lecture Count i.e. attendance data should exist and be available**
Pre-condition(s)	**Student is enrolled in >= 1 course**
Post-condition(s)	**Total recorded tardies are displayed**

Typical Course of Action		
S #	**Actor Action**	**System Response**
1	**Student goes to Attendance Section**	**System fetches enrolled courses**
2	**Student selects course to view**	**System retrieves the attendance data for that specific course**
3		**System displays total tardies**

Alternate Course of Action		
S #	**Actor Action**	**System Response**
1	**No tardies have been recorded**	**System displayed relevant error message**
2	**No lectures have been held yet**	**System displays relevant error message**

Identifier	**UC-23**
Name	**Manage Student Records**
Summary	**An admin views, edits, or removes student academic and personal records stored**
Priority	**High**

Actors	**Admin, Database**
Dependencies	**Depends on Role Based Access i.e. the admin must be logged in**
Pre-condition(s)	**Admin is logged in**
Post-condition(s)	**The selected record is updated**

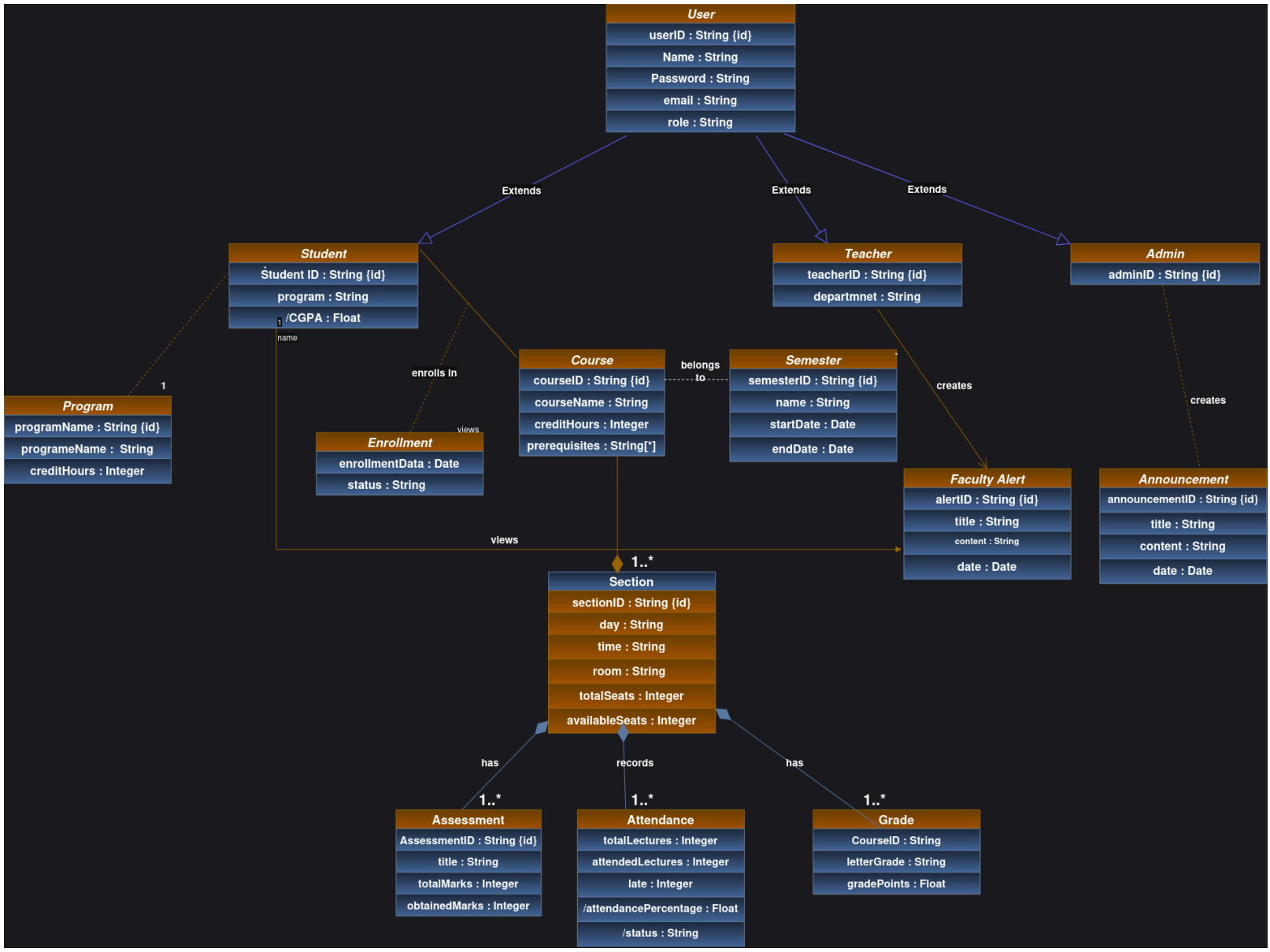
Typical Course of Action		
S #	**Actor Action**	**System Response**
1	**Admin navigates to the Student Records**	**System displays a searchable list of all registered students**
2	**Admin searches for student**	**System filters and displays the matching student record**
3	**Admin makes necessary changes and saves**	**System validates and updates the record in the Database**

Alternate Course of Action		
S #	**Actor Action**	**System Response**
1	**Admin searches for a student that does not exist**	**System displays relevant error message**

Identifier	**UC-24**
Name	**Manage Course Listings**
Summary	**An admin creates, updates, or removes course**

	listings available for student enrollment
Priority	**High**
Actors	**Admin**
Dependencies	**Depends on Role Based Access i.e. the admin must be logged in**
Pre-condition(s)	**Admin is logged in**
Post-condition(s)	**The course catalog is updated and the changes are reflected in the student-facing section**

Typical Course of Action		
S #	**Actor Action**	**System Response**
1	**Admin navigates to the Course Catalog Management section**	**System displays existing course listings**
2	**Admin selects to add a new course or edit an existing one**	**System displays the course creation or editing form**
3	**Admin fills in details and saves**	**System validates fields and inputs them into database**

Alternate Course of Action		
S #	**Actor Action**	**System Response**
1	**Admin attempts to add a course with a duplicate course code**	**System displays relevant error message**
2	**Admin attempts to delete a course with active enrollments**	**System warns the admin and requires confirmation**

ANALYSIS CLASS DIAGRAM